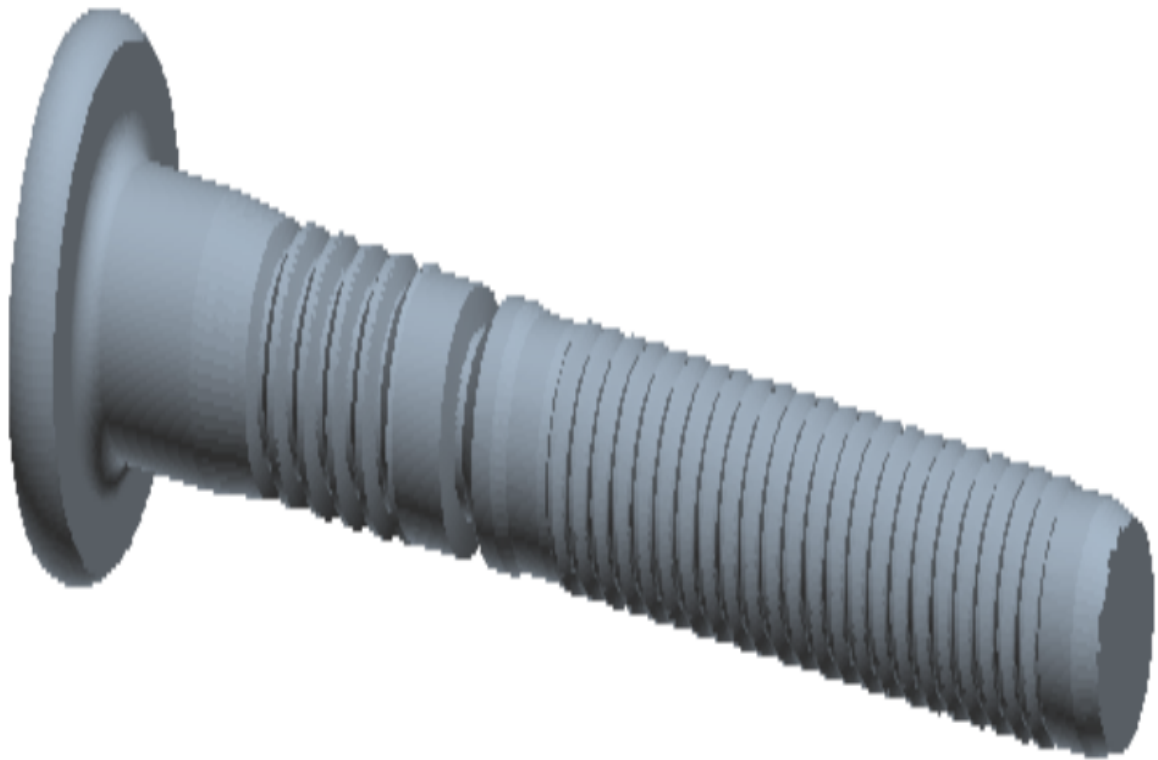
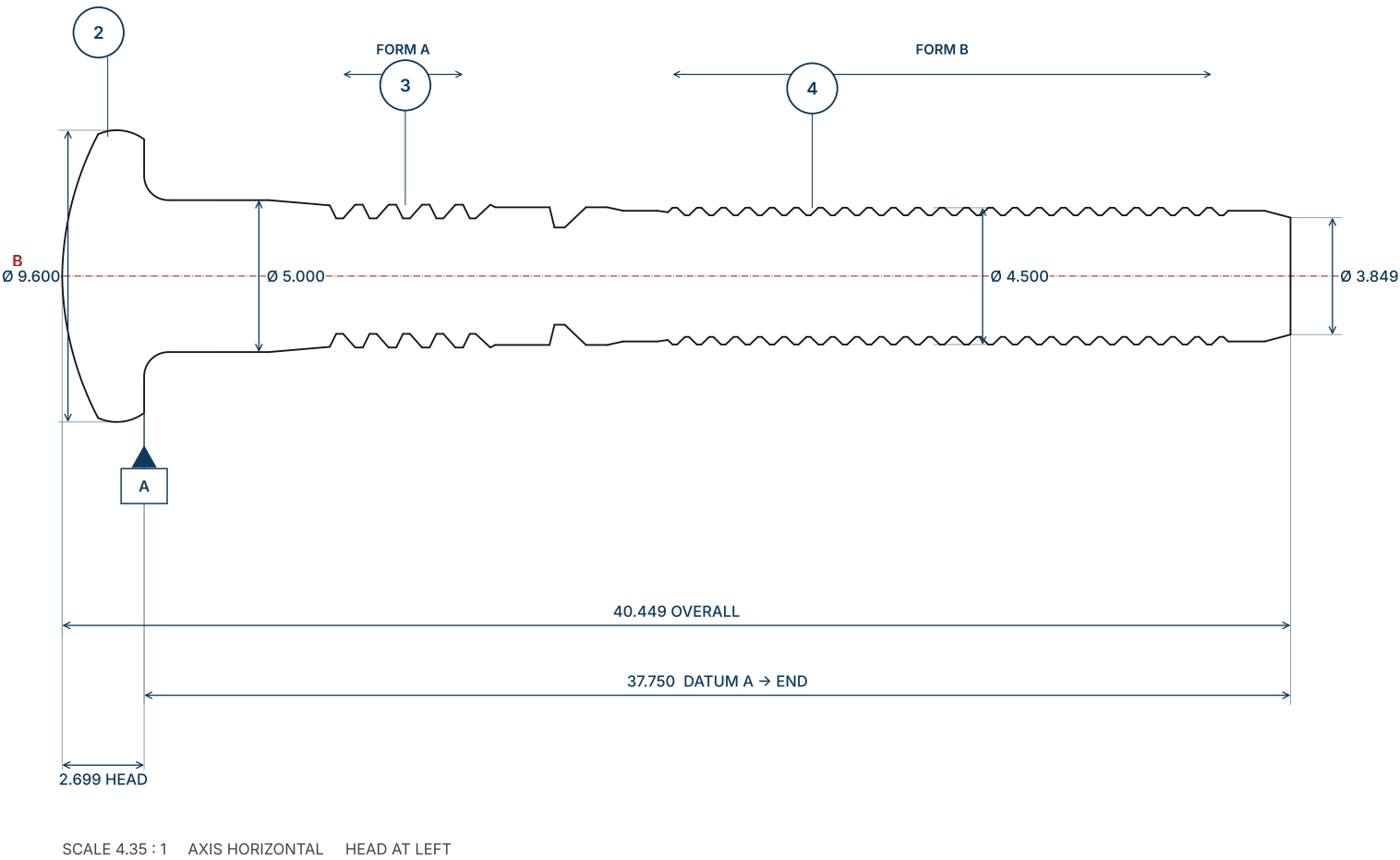


Solid of revolution about datum B. Revolve the axial profile.

Sizes are CAD nominals from 4.8_LB.STEP.

No tolerance, material, or finish is stored in the model.

Graphic sizes are rounded to 0.001 mm.

Sheet 6 lists every profile segment to 0.00001 mm.

Positive Z runs from datum A toward the shank end.

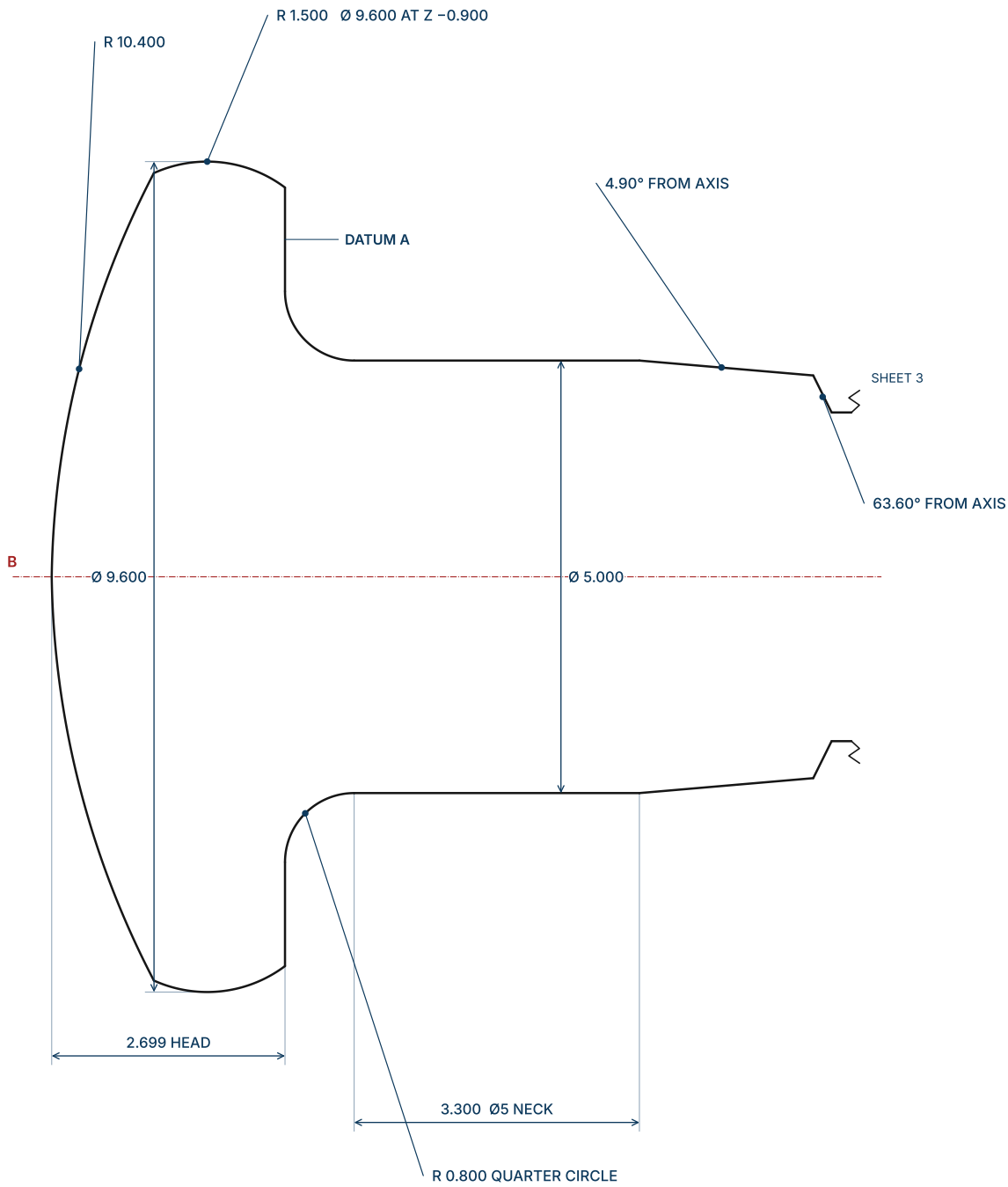
Balloons 2, 3 and 4 mark the detail sheets.

Form A pitch 1.100. Form B pitch 0.800, 23 crests.



PART	4.8 LB	SHEET	1 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0	DATUM B AXIS		

DETAIL 2 SCALE 13 : 1 FULL PROFILE



Nose arc center is 0.150 mm past datum B, at Z = 7.700.

R1.500 meets the nose at a corner. R0.800 is tangent

to datum A and to the Ø5.000 neck.

Datum A is the annular face from Ø6.600 to Ø9.000.



HEAD STATIONS FROM DATUM A

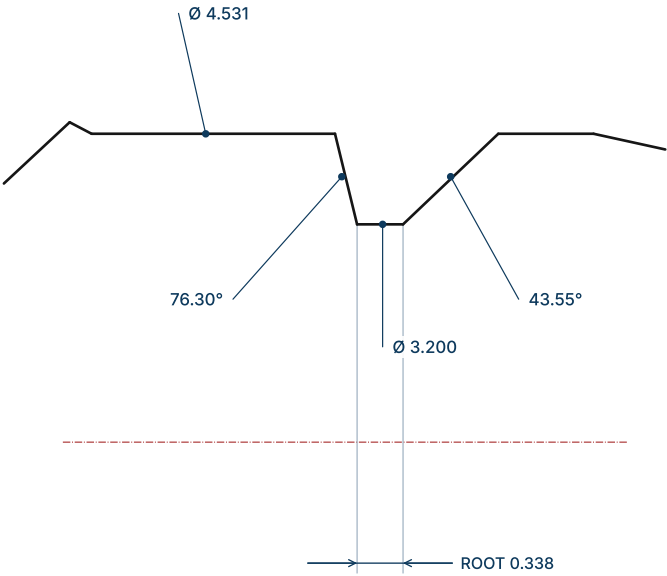
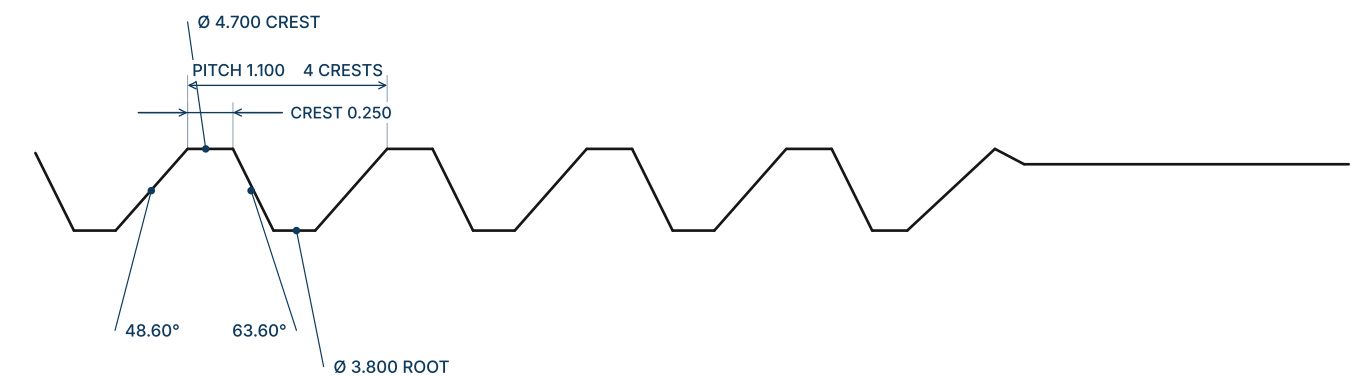
PT	Z	RADIUS	Ø	FEATURE
1	-2.69892	0	0	TIP ON AXIS
2	-1.51709	4.66719	9.334	NOSE / HEAD ARC JOIN
3	-0.90000	4.80000	9.600	LARGEST DIAMETER
4	0.00000	4.50000	9.000	DATUM A, OUTER
5	0.00000	3.30000	6.600	DATUM A, INNER
6	0.80000	2.50000	5.000	NECK START, R0.8 END
7	4.10000	2.50000	5.000	NECK END
8	6.11113	2.32759	4.655	4.90° TAPER END
9	6.32338	1.90000	3.800	63.60° END, Ø3.800
10	6.55327	1.90000	3.800	FORM A STARTS

AXIAL-SECTION ARCS

ARC	R	CENTER R	CENTER Z	NOTE
NOSE	10.40000	-0.15000	7.70000	TORUS, CENTER PAST THE AXIS
HEAD	1.50000	3.30000	-0.90000	THROUGH Ø9.600
FILLET	0.80000	3.30000	0.80000	QUARTER CIRCLE, TANGENT

PART	4.8 LB	SHEET	2 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0	DATUM B AXIS		

DETAIL 3 FORM A SCALE 24 : 1 GROOVE SCALE 18 : 1 HALF PROFILE ABOVE THE AXIS



FORM A — 4 CRESTS, PITCH 1.100

Crest Ø 4.700, flat 0.250. Root Ø 3.800, flat 0.230.

Rising flank 48.60° from the axis. Falling flank 63.60°.

First rising flank starts at Z of station 10 on sheet 2.

#	CREST Z0	CREST Z1	FLAT	NEXT
1	6.95000	7.20000	0.250	8.05000
2	8.05000	8.30000	0.250	9.15000
3	9.15000	9.40000	0.250	10.25000
4	10.25000	10.50000	0.250	RUNOUT

FORM A RUNOUT

STEP	Z	RADIUS	Ø	NOTE
SHORT ROOT	10.91659	1.90000	3.800	AFTER 4th FALLING FLANK
42.95° T0	11.40000	2.35000	4.700	NO CREST FLAT
27.70° T0	11.56107	2.26544	4.531	BARREL START

GROOVE

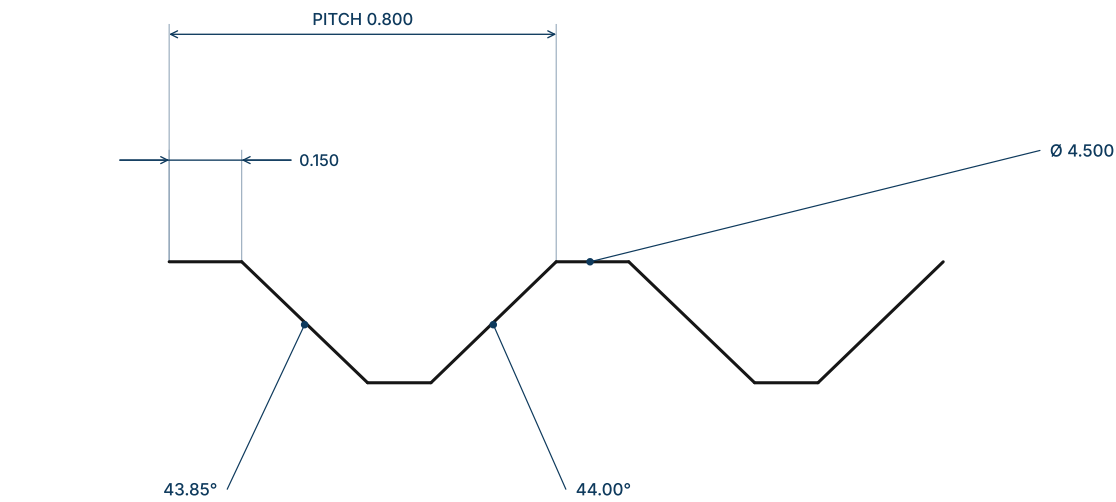
FEATURE	Z0	Z1	LENGTH	Ø
BARREL	11.56107	13.35000	1.789	4.531
76.30° FLANK	13.35000	13.51222	0.162	→ Ø3.200
GROOVE ROOT	13.51222	13.85000	0.338	3.200
43.55° FLANK	13.85000	14.55000	0.700	→ Ø4.531
SHORT BARREL	14.55000	15.25000	0.700	4.531



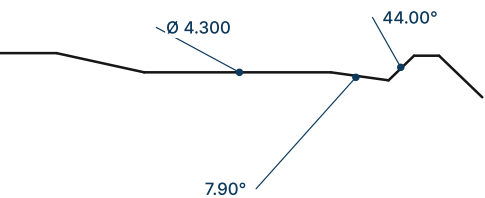
3RD ANGLE

PART	4.8 LB	SHEET	3 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0	DATUM B AXIS		

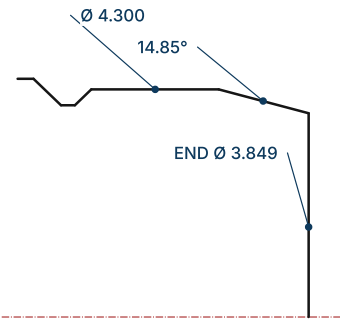
DETAIL 4 THREAD ENTRY AND TIP ARE TRUE RADIAL SCALE TOOTH DETAIL IS ENLARGED



TOOTH SCALE 64 : 1 ROOT Ø 4.000 MATERIAL IS BELOW THE LINE 23 CRESTS



THREAD ENTRY SCALE 22 : 1



TIP SCALE 14 : 1



FORM B CREST STATIONS — ALL 23

#	Z0	Z1	#	Z0	Z1
1	17.40000	17.55000	13	27.00000	27.15000
2	18.20000	18.35000	14	27.80000	27.95000
3	19.00000	19.15000	15	28.60000	28.75000
4	19.80000	19.95000	16	29.40000	29.55000
5	20.60000	20.75000	17	30.20000	30.35000
6	21.40000	21.55000	18	31.00000	31.15000
7	22.20000	22.35000	19	31.80000	31.95000
8	23.00000	23.15000	20	32.60000	32.75000
9	23.80000	23.95000	21	33.40000	33.55000
10	24.60000	24.75000	22	34.20000	34.35000
11	25.40000	25.55000	23	35.00000	35.15000
12	26.20000	26.35000			

FORM B DEFINITION

ITEM	VALUE	NOTE
PITCH	0.800	CREST START TO CREST START
CREST Ø	4.500	AXIAL FLAT 0.150
ROOT Ø	4.000	AXIAL FLAT 0.131
FALLING FLANK	43.85°	FROM THE AXIS, CREST TO ROOT
RISING FLANK	44.00°	FROM THE AXIS, ROOT TO CREST
COUNT	23	FIRST Z0 17.40000 LAST Z1 35.15000
AFTER LAST ROOT	35.69645	44.00° STOPS AT Ø 4.300

PART	4.8 LB	SHEET	4 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0		DATUM B AXIS	

ALL CONTROLLING SIZES. Z IS FROM DATUM A, POSITIVE TOWARD THE SHANK END. ANGLES ARE FROM THE AXIS.

1 OVERALL

ITEM	0.001 mm	MODEL	NOTE
OVERALL LENGTH	40.449	40.44892	TIP TO END FACE
TIP Z	-2.699	-2.69892	ON THE AXIS
END FACE Z	37.750	37.75000	FULL FACE TO THE AXIS
LARGEST Ø	9.600	9.60000	AT Z = -0.900
END FACE Ø	3.849	3.84925	SHANK END
VOLUME	718.326 mm³	718.32579	SOLID, NO BORE
SURFACE AREA	792.966 mm²	792.966	
CENTRE OF MASS Z	14.024	14.02383	ON THE AXIS

3 ARCS IN THE AXIAL SECTION

ARC	R	CENTER R	CENTER Z	ARC LENGTH
NOSE	10.40000	-0.15000	7.70000	4.85856
HEAD	1.50000	3.30000	-0.90000	1.60122
FILLET	0.80000	3.30000	0.80000	1.25664

5 FORM B AND TIP

ITEM	VALUE	NOTE
B PITCH	0.80000	23 CRESTS
B CREST	Ø 4.500 FLAT 0.150	Z 17.40000 FIRST
B ROOT	Ø 4.000 FLAT 0.131	BETWEEN FLANKS
B FLANKS	43.85° AND 44.00°	FROM THE AXIS
B LAST CREST END	35.15000	Z
B RUNOUT	Ø 4.300 AT Z 35.69645	44.00° PARTIAL FLANK
TIP LAND	Ø 4.300 L 1.204	TO Z 36.90000
TIP TAPER	14.85°	TO Ø 3.849
END FACE	Z 37.75000	Ø 3.849 TO THE AXIS

2 HEAD AND NECK

POINT	Z	RADIUS	Ø	NOTE
TIP	-2.69892	0.00000	0.000	R10.400 START
ARC JOIN	-1.51709	4.66719	9.334	NOSE TO R1.500
MAX Ø	-0.90000	4.80000	9.600	ON THE R1.500 ARC
DATUM A OUT	0.00000	4.50000	9.000	SHOULDER OUTER
DATUM A IN	0.00000	3.30000	6.600	SHOULDER INNER
NECK START	0.80000	2.50000	5.000	R0.800 END
NECK END	4.10000	2.50000	5.000	LENGTH 3.300
TAPER END	6.11113	2.32759	4.655	4.90°
CHAMFER END	6.32338	1.90000	3.800	63.60°
FORM A START	6.55327	1.90000	3.800	LEAD-IN LAND END

4 FORM A, GROOVE, ENTRY

ITEM	VALUE	WHERE
A CREST Ø / FLAT	4.700 / 0.250	4 CRESTS
A ROOT Ø / FLAT	3.800 / 0.230	PITCH 1.100
A FLANKS	48.60° / 63.60°	RISE / FALL, FROM AXIS
A FIRST CREST	6.95000	Z0
A LAST CREST END	10.50000	Z1
BARREL Ø	4.531	Z 11.56107 → 13.35000
GROOVE Ø	3.200	FLAT 0.338 AT Z 13.51222
GROOVE FLANKS	76.30° / 43.55°	INTO / OUT OF Ø3.200
SHORT BARREL	4.531	LENGTH 0.700
Ø4.300 LAND	4.300	Z 15.77723 → 16.90000
ENTRY TAPERS	12.35° THEN 7.90°	ONTO THE PRE-THREAD Ø
PRE-THREAD Ø	4.204	Z 17.24664



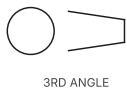
ANGLES ARE THE MODEL VALUES. A 0.001 mm CALLOUT IS THE MODEL SIZE ROUNDED TO THE NEAREST MICROMETRE.

PART	4.8 LB	SHEET	5 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	

EVERY SEGMENT OF THE +Z AXIAL PROFILE, TIP TO END FACE. REVOLVE ABOUT DATUM B. VALUES IN MILLIMETRES TO 0.00001.

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
001	ARC	-2.69892	0.00000	-1.51709	4.66719	4.85856		R10.40000
002	ARC	-1.51709	4.66719	0.00000	4.50000	1.60122		R1.50000
003	FACE	0.00000	4.50000	0.00000	3.30000	1.20000	90.00°	
004	ARC	0.00000	3.30000	0.00000	2.50000	1.25664		R0.80000
005	CYL	0.80000	2.50000	4.10000	2.50000	3.30000	0.00°	
006	TAPER	4.10000	2.50000	6.11113	2.32759	2.01850	4.90°	
007	TAPER	6.11113	2.32759	6.32338	0.47737	1.90000	63.60°	
008	CYL	6.32338	1.90000	6.55327	1.90000	0.22989	0.00°	
009	TAPER	6.55327	1.90000	6.95000	2.35000	0.59991	48.60°	
010	CYL	6.95000	2.35000	7.20000	2.35000	0.25000	0.00°	
011	TAPER	7.20000	2.35000	7.42338	1.90000	0.50239	63.60°	
012	CYL	7.42338	1.90000	7.65327	1.90000	0.22989	0.00°	
013	TAPER	7.65327	1.90000	8.05000	2.35000	0.59991	48.60°	
014	CYL	8.05000	2.35000	8.30000	2.35000	0.25000	0.00°	
015	TAPER	8.30000	2.35000	8.52338	1.90000	0.50239	63.60°	
016	CYL	8.52338	1.90000	8.75327	1.90000	0.22989	0.00°	
017	TAPER	8.75327	1.90000	9.15000	2.35000	0.59991	48.60°	
018	CYL	9.15000	2.35000	9.40000	2.35000	0.25000	0.00°	
019	TAPER	9.40000	2.35000	9.62338	1.90000	0.50239	63.60°	
020	CYL	9.62338	1.90000	9.85327	1.90000	0.22989	0.00°	
021	TAPER	9.85327	1.90000	10.25000	2.35000	0.59991	48.60°	
022	CYL	10.25000	2.35000	10.50000	2.35000	0.25000	0.00°	
023	TAPER	10.50000	2.35000	10.72338	1.90000	0.50239	63.60°	
024	CYL	10.72338	1.90000	10.91659	1.90000	0.19321	0.00°	
025	TAPER	10.91659	1.90000	11.40000	2.35000	0.66044	42.95°	
026	TAPER	11.40000	2.35000	11.56107	2.26544	0.18192	27.70°	
027	CYL	11.56107	2.26544	13.35000	2.26544	1.78093	0.00°	
028	TAPER	13.35000	2.26544	13.51222	1.60000	0.68492	76.30°	
029	CYL	13.51222	1.60000	13.85000	1.60000	0.33778	0.00°	
030	TAPER	13.85000	1.60000	14.55000	2.26544	0.96502	43.55°	
031	CYL	14.55000	2.26544	15.25000	2.26544	0.70000	0.00°	
032	TAPER	15.25000	2.26544	15.77723	2.15000	0.53972	12.35°	
033	CYL	15.77723	2.15000	16.90000	2.15000	1.12277	0.00°	
034	TAPER	16.90000	2.15000	17.24664	2.10190	0.34996	7.90°	
035	TAPER	17.24664	2.10190	17.40000	2.25000	0.21320	44.00°	
036	CYL	17.40000	2.25000	17.55000	2.25000	0.15000	0.00°	
037	TAPER	17.55000	2.25000	17.81024	2.00000	0.36007	43.85°	
038	CYL	17.81024	2.00000	17.94112	2.00000	0.13007	0.00°	
039	TAPER	17.94112	2.00000	18.20000	2.25000	0.35989	44.00°	
040	CYL	18.20000	2.25000	18.35000	2.25000	0.15000	0.00°	
041	TAPER	18.35000	2.25000	18.61024	2.00000	0.36007	43.85°	
042	CYL	18.61024	2.00000	18.74112	2.00000	0.13007	0.00°	
043	TAPER	18.74112	2.00000	19.00000	2.25000	0.35989	44.00°	
044	CYL	19.00000	2.25000	19.15000	2.25000	0.15000	0.00°	
045	TAPER	19.15000	2.25000	19.41024	2.00000	0.36007	43.85°	
046	CYL	19.41024	2.00000	19.54112	2.00000	0.13007	0.00°	
047	TAPER	19.54112	2.00000	19.80000	2.25000	0.35989	44.00°	
048	CYL	19.80000	2.25000	19.95000	2.25000	0.15000	0.00°	

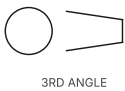
#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
049	TAPER	19.95000	2.25000	20.21024	2.00000	0.36007	43.85°	
050	CYL	20.21024	2.00000	20.34112	2.00000	0.13007	0.00°	
051	TAPER	20.34112	2.00000	20.60000	2.25000	0.35989	44.00°	
052	CYL	20.60000	2.25000	20.75000	2.25000	0.15000	0.00°	
053	TAPER	20.75000	2.25000	21.01024	2.00000	0.36007	43.85°	
054	CYL	21.01024	2.00000	21.14112	2.00000	0.13007	0.00°	
055	TAPER	21.14112	2.00000	21.40000	2.25000	0.35989	44.00°	
056	CYL	21.40000	2.25000	21.55000	2.25000	0.15000	0.00°	
057	TAPER	21.55000	2.25000	21.81024	2.00000	0.36007	43.85°	
058	CYL	21.81024	2.00000	21.94112	2.00000	0.13007	0.00°	
059	TAPER	21.94112	2.00000	22.20000	2.25000	0.35989	44.00°	
060	CYL	22.20000	2.25000	22.35000	2.25000	0.15000	0.00°	
061	TAPER	22.35000	2.25000	22.61024	2.00000	0.36007	43.85°	
062	CYL	22.61024	2.00000	22.74112	2.00000	0.13007	0.00°	
063	TAPER	22.74112	2.00000	23.00000	2.25000	0.35989	44.00°	
064	CYL	23.00000	2.25000	23.15000	2.25000	0.15000	0.00°	
065	TAPER	23.15000	2.25000	23.41024	2.00000	0.36007	43.85°	
066	CYL	23.41024	2.00000	23.54112	2.00000	0.13007	0.00°	
067	TAPER	23.54112	2.00000	23.80000	2.25000	0.35989	44.00°	
068	CYL	23.80000	2.25000	23.95000	2.25000	0.15000	0.00°	
069	TAPER	23.95000	2.25000	24.21024	2.00000	0.36007	43.85°	
070	CYL	24.21024	2.00000	24.34112	2.00000	0.13007	0.00°	
071	TAPER	24.34112	2.00000	24.60000	2.25000	0.35989	44.00°	
072	CYL	24.60000	2.25000	24.75000	2.25000	0.15000	0.00°	
073	TAPER	24.75000	2.25000	25.01024	2.00000	0.36007	43.85°	
074	CYL	25.01024	2.00000	25.14112	2.00000	0.13007	0.00°	
075	TAPER	25.14112	2.00000	25.40000	2.25000	0.35989	44.00°	
076	CYL	25.40000	2.25000	25.55000	2.25000	0.15000	0.00°	
077	TAPER	25.55000	2.25000	25.81024	2.00000	0.36007	43.85°	
078	CYL	25.81024	2.00000	25.94112	2.00000	0.13007	0.00°	
079	TAPER	25.94112	2.00000	26.20000	2.25000	0.35989	44.00°	
080	CYL	26.20000	2.25000	26.35000	2.25000	0.15000	0.00°	
081	TAPER	26.35000	2.25000	26.61024	2.00000	0.36007	43.85°	
082	CYL	26.61024	2.00000	26.74112	2.00000	0.13007	0.00°	
083	TAPER	26.74112	2.00000	27.00000	2.25000	0.35989	44.00°	
084	CYL	27.00000	2.25000	27.15000	2.25000	0.15000	0.00°	
085	TAPER	27.15000	2.25000	27.41024	2.00000	0.36007	43.85°	
086	CYL	27.41024	2.00000	27.54112	2.00000	0.13007	0.00°	
087	TAPER	27.54112	2.00000	27.80000	2.25000	0.35989	44.00°	
088	CYL	27.80000	2.25000	27.95000	2.25000	0.15000	0.00°	
089	TAPER	27.95000	2.25000	28.21024	2.00000	0.36007	43.85°	
090	CYL	28.21024	2.00000	28.34112	2.00000	0.13007	0.00°	
091	TAPER	28.34112	2.00000	28.60000	2.25000	0.35989	44.00°	
092	CYL	28.60000	2.25000	28.75000	2.25000	0.15000	0.00°	
093	TAPER	28.75000	2.25000	29.01024	2.00000	0.36007	43.85°	
094	CYL	29.01024	2.00000	29.14112	2.00000	0.13007	0.00°	
095	TAPER	29.14112	2.00000	29.40000	2.25000	0.35989	44.00°	
096	CYL	29.40000	2.25000	29.55000	2.25000	0.15000	0.00°	



PART	4.8 LB	SHEET	6 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0		DATUM B AXIS	

EVERY SEGMENT OF THE +Z AXIAL PROFILE, TIP TO END FACE. REVOLVE ABOUT DATUM B. VALUES IN MILLIMETRES TO 0.00001.

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
097	TAPER	29.55000	2.25000	29.81024	2.00000	0.36087	43.85°	
098	CYL	29.81024	2.00000	29.94112	2.00000	0.13087	0.00°	
099	TAPER	29.94112	2.00000	30.20000	2.25000	0.35989	44.00°	
100	CYL	30.20000	2.25000	30.35000	2.25000	0.15000	0.00°	
101	TAPER	30.35000	2.25000	30.61024	2.00000	0.36087	43.85°	
102	CYL	30.61024	2.00000	30.74112	2.00000	0.13087	0.00°	
103	TAPER	30.74112	2.00000	31.00000	2.25000	0.35989	44.00°	
104	CYL	31.00000	2.25000	31.15000	2.25000	0.15000	0.00°	
105	TAPER	31.15000	2.25000	31.41024	2.00000	0.36087	43.85°	
106	CYL	31.41024	2.00000	31.54112	2.00000	0.13087	0.00°	
107	TAPER	31.54112	2.00000	31.80000	2.25000	0.35989	44.00°	
108	CYL	31.80000	2.25000	31.95000	2.25000	0.15000	0.00°	
109	TAPER	31.95000	2.25000	32.21024	2.00000	0.36087	43.85°	
110	CYL	32.21024	2.00000	32.34112	2.00000	0.13087	0.00°	
111	TAPER	32.34112	2.00000	32.60000	2.25000	0.35989	44.00°	
112	CYL	32.60000	2.25000	32.75000	2.25000	0.15000	0.00°	
113	TAPER	32.75000	2.25000	33.01024	2.00000	0.36087	43.85°	
114	CYL	33.01024	2.00000	33.14112	2.00000	0.13087	0.00°	
115	TAPER	33.14112	2.00000	33.40000	2.25000	0.35989	44.00°	
116	CYL	33.40000	2.25000	33.55000	2.25000	0.15000	0.00°	
117	TAPER	33.55000	2.25000	33.81024	2.00000	0.36087	43.85°	
118	CYL	33.81024	2.00000	33.94112	2.00000	0.13087	0.00°	
119	TAPER	33.94112	2.00000	34.20000	2.25000	0.35989	44.00°	
120	CYL	34.20000	2.25000	34.35000	2.25000	0.15000	0.00°	
121	TAPER	34.35000	2.25000	34.61024	2.00000	0.36087	43.85°	
122	CYL	34.61024	2.00000	34.74112	2.00000	0.13087	0.00°	
123	TAPER	34.74112	2.00000	35.00000	2.25000	0.35989	44.00°	
124	CYL	35.00000	2.25000	35.15000	2.25000	0.15000	0.00°	
125	TAPER	35.15000	2.25000	35.41024	2.00000	0.36087	43.85°	
126	CYL	35.41024	2.00000	35.54112	2.00000	0.13087	0.00°	
127	TAPER	35.54112	2.00000	35.69645	2.15000	0.21593	44.00°	
128	CYL	35.69645	2.15000	36.90000	2.15000	1.20355	0.00°	
129	TAPER	36.90000	2.15000	37.75000	1.92463	0.87937	14.85°	
130	FACE	37.75000	1.92463	37.75000	0.00000	1.92463	90.00°	



PART	4.8 LB	SHEET	7 / 7
NAME	SMALL NUT	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0	DATUM B AXIS		